Global Coffee Sustainability Dashboard – SDG 12 (Responsible Consumption & Production)

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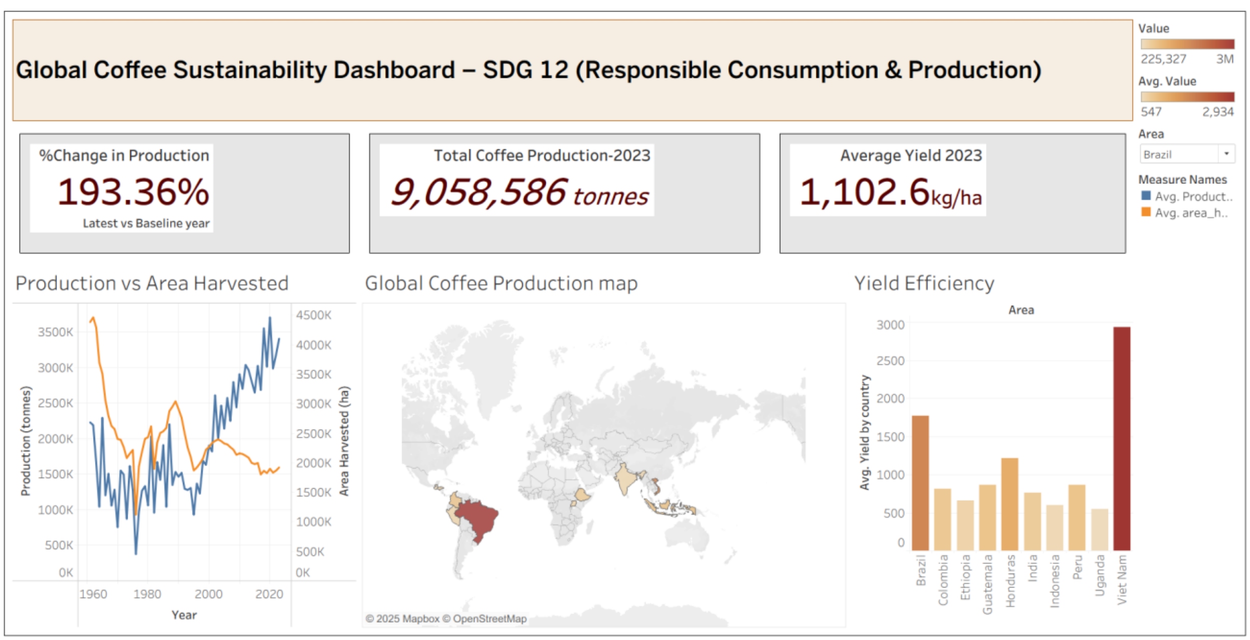
1.Introduction

This is the business intelligence case study of the coffee production using tableau dashboard created with the help of FAOSTAT data (1961-2023). Coffee is one the biggest significant agricultural commodity and it's a huge part of supply chain of multinational firms such as Starbucks. Due to the rise of the global demand, production trends, land-usage, and sustainability impacts becomes important for companies aligning with SDG 12 – Responsible Consumption and Production.

Through this analysis we used business intelligence to extract real world agricultural insights and identify sustainability implications relevant to corporate sourcing strategies. This dashboard includes multiple visuals including, KPI cards, a dual axis chart, a geographical map, and a yield comparison chart- to analyze production growth, resource pressure, and country level analysis. This case study report provides strategic recommendations for Starbucks to improve the sustainability of their global supply chains.

2.Data and dashboard overview

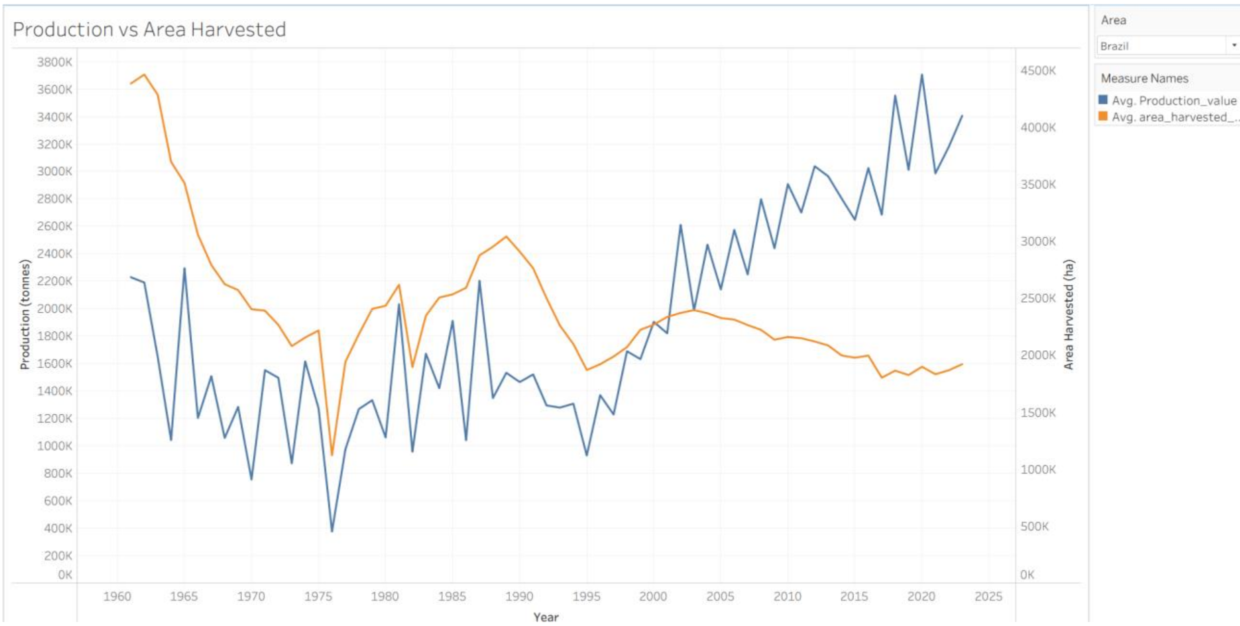


This dashboard is created using a cleaned dataset acquired from FAOSTAT, it contains country-level analysis including Brazil, Vietnam, Colombia, Indonesia, Ethiopia, Honduras, Uganda, Peru, India and Guatemala. These 10 countries contribute the highest global coffee output, which makes them ideal for the analysis of sustainability performance. This dashboard contains three KPI cards which summarizes some important global sustainability indicators such as total production for 2023, average yield which represents productivity levels, and percentage growth since baseline year.

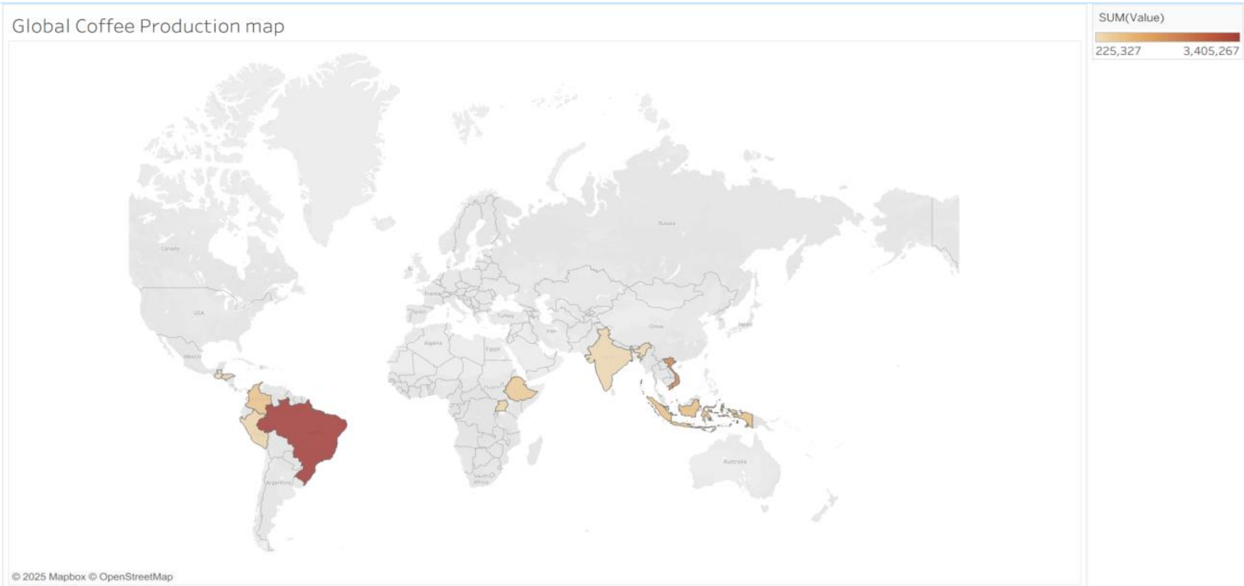
This dashboard also includes charts like dual-axis line graph comparing how production and harvested land have changed over time, a geographic map that visualizes the production distribution among different countries, and a yield comparison that highlights efficiency differences. All the visuals and KPIs illustrate a coherent, multi-layered narrative that helps in decision-making for sustainable supply chain.

3.Dashboard Insights

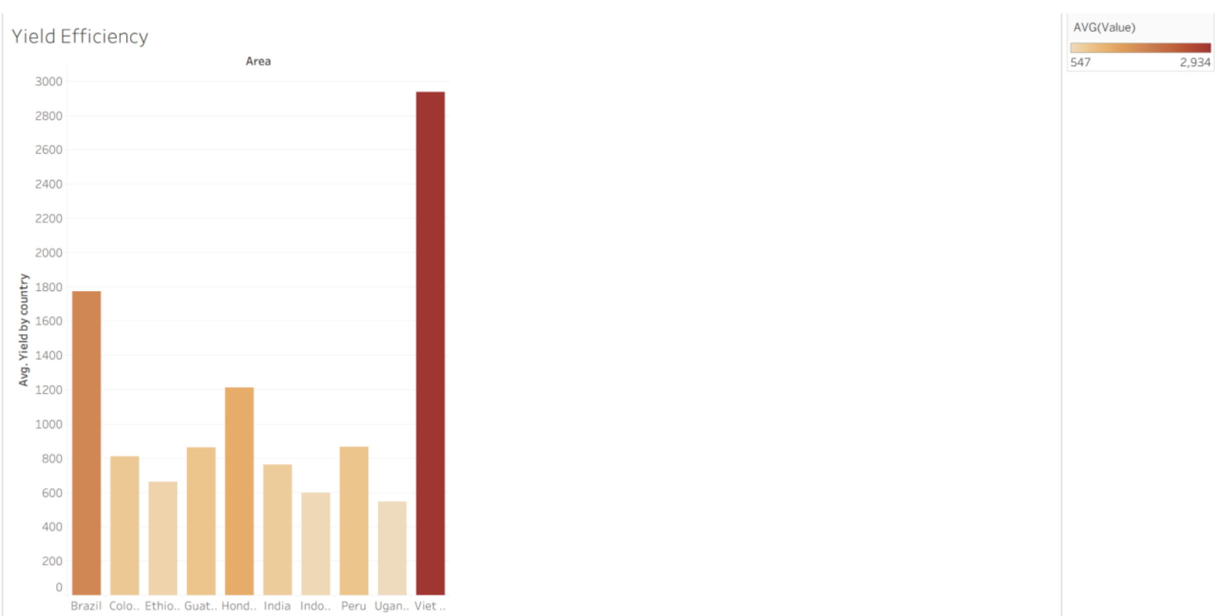
Through the KPIs we can get an overview of the global coffee sustainability performance. The first KPI shows that there has been an increase of 193.36% in the global coffee production since 1961, that shows the rapid expansion of coffee consumption worldwide. This insight is important for companies like Starbucks to confirm that the demand has expanded aligning with the global supply. Through the second KPI we can see the coffee production in 2023 that is 9,058,586 tonnes that highlights the scale of the coffee industry. It also tells us about the international coffee supply depends on the agricultural regions located within the tropics. The final KPI shows us the average yield which relates to the sustainability, as higher yields represent more efficient use of land. The average yield is 1,102.6 kg/ha that tells us how much efficient land is used per kg of coffee.



The dual-axis line chart provides a deeper insight into the relationship between production vs area harvested. In countries like Brazil, the production has increased over the years while the area harvested has remained relatively stable. This shows that through improved technology, agronomy and crop management we can increase the output without increasing the area harvested. This pattern highly with the SDG 12 goals by demonstrating production efficiency and reduced environmental pressure. Whereas in countries like Ethiopia, Honduras and Uganda, increase in production lead to increase in area harvested. Land expansion based growth leads to certain risks as deforestation, soil degradation and biodiversity loss. Starbucks can use these insights to target areas where there is a need of improving productivity without being over dependent on expanding land, its will highlight Starbucks contribution towards sustainability.



This geographic map shows us that there is a large contribution in the global coffee production in countries like Brazil, followed by Vietnam, Cambodia and Indonesia. Due to so much production in only a few countries, the global supply chain becomes more vulnerable to risks such as climatic changes, bad weather, crop diseases and political issues. In African countries, there are smaller producers but they are emerging, which offers opportunities that could help in expanding supply in the future. Starbucks can use these insights to see which regions carry more risks and where investments can be made to build new partnerships.



The yield comparison bar graph shows us the major differences in efficient farming across countries. Vietnam stands out as the most efficient producer amongst all the other countries. It is achieving yields above 2,800 hg/ha, which reflects high-intensity farming systems, government support and impressive agricultural infrastructure. Brazil also shows efficient production of coffee achieving above average yields. Whereas in countries like Ethiopia, Uganda and Peru the yields are below average which shows their usage of traditional farming, dispersed producers, and limited agronomic support. These insights might be helpful for Starbucks to understand which supply regions may struggle to meet long-term demands due to lack of efficient farming methods and increasing land expansion which directly undermines SDG 12.

4.Recommendations

Based on the dashboard Starbucks can take several decisions that can help them to align with the SDG 12 goals.

Firstly, Starbucks should focus on helping the countries like Ethiopia, Uganda and Peru to increase their yields efficiently by training the farmers, provide them with better seedlings, etc. Through these steps we could improve the production without expanding land area, that helps in reducing environmental pressure.

Secondly, the company should build long-term partnership with suppliers in regions that are less efficient. Offering them stable contracts, paying sustainability premiums and supporting community projects will help farmers to be more efficient.

Thirdly, Starbucks can focus on diversifying its suppliers. Brazil and Vietnam produces coffee efficiently but depending too much on them makes the global supply chain vulnerable. Sourcing from emerging regions like East Africa would make global supply more stable.

Practices like soil restoration, improved biodiversity and decreasing carbon emissions should be focused on by Starbucks to contribute in its goal of sustainability.

5.Ethical Consideration and BI reflection

The BI process shows how raw data can be interpreted carefully to guide sustainable decisions. While FAOSTAT is reliable, it lacks data on social and environmental factors. Ethical BI does not mean looking at only numbers, as high yields do not guarantee sustainable practices and low yield maybe caused due to social or economic challenges. BI insights should be used in alignment of sustainability.

6.Conclusion

This dashboard gave a clear picture of global coffee production and sustainability issues around it. It shows long-term growth, differences in yield efficiency and heavy reliance on a few major producers. These insights highlight the involvement of Starbucks to improve sustainability by supporting low efficiency regions and better diversification. Using these insights along with the SDG 12 goals will help build a more resilient and sustainable supply chain.

7.References

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